Information Geometry of Classical Carriers and Non-Classical Payloads in Biological Information Processing

Ian Todd
Sydney Medical School
University of Sydney
Sydney, Australia

Abstract

The central claim of this paper is that dimensionality in biological information processing need not live in carrier frequency. Decoherence maps a d-level quantum system from a (d^2-1)-dimensional Bures state manifold onto a $(d-1)$-dimensional Fisher–Rao simplex, annihilating the d^2-d coherence directions. This dimensional collapse supplies a continuous non-classicality index χ from Bures principal angles, allowing one to quantify not only whether a system is classical or quantum, but how much non-classical structure remains geometrically accessible. We use that geometry to distinguish *classical carriers* from *non-classical payloads*. Photosynthetic exciton transport provides the proof of principle: validated FMO and PE545 models operate near the quantum-classical boundary, showing that biologically functional dynamics can live in a near-classical but non-classical regime. Neural oscillations, by contrast, are deep classical at carrier scale. A minimal KcsA-like ion-channel payload calculation then gives a concrete neural-side anchor: at $\gamma_{\text{bio}}/J_{\max} = 6.67$, the biological point remains boundary-near with $\theta_{\min} \approx 88.55°$ and $\chi \approx 2.76 \times 10^{-4}$. We then show that deep-classical carriers do not force low-dimensionality. For molecular modules i with local non-classical load $L_i = \chi_i(d_i^2 - d_i)$, the redundancy-adjusted payload $\mathcal{L}_{\text{eff}}(E) = \varrho_E^{-1} \sum_{i \in E} L_i$ is extensive whenever a nonzero fraction of modules sit near the boundary and redundancy is bounded. Under a minimal phase-window carrier model, coordinated dimensionality grows as the payload coordination radius expands and saturates at an amplitude ceiling: lower-frequency carriers can therefore support larger coordinated dimensionality not because the carrier itself is quantum, but because finite propagation and phase tolerance allow them to entrain larger sets of molecular payload modules while adding little geometric mixing of their own. This reframes the usual high-frequency/high-dimensional intuition: frequency is a coordination scale, not the locus of dimensionality. The resulting picture is a multi-scale architecture in which deep-classical macroscopic waves coordinate an extensive near-boundary molecular substrate.

1 Introduction

High-dimensional neural activity is often discussed as if it were primarily a property of fast dynamics. In the working-memory literature, gamma and beta bursts are closely associated with access, gating, and mixed selectivity [1, 2, 3, 4]. Read one way, that literature supports a simple heuristic: high frequency goes with high dimensionality. This paper argues that the heuristic places dimensionality in the wrong object. Frequency is a temporal property of a carrier. Dimensionality is a property of a state space. Those are not the same thing.

This paper develops a different answer. The claim is not that low-frequency oscillations are themselves quantum, nor that high dimensionality automatically implies non-classicality. The

claim is narrower and stronger: biological systems are plausibly organized as *carrier-payload architectures*. At macroscopic scale, oscillatory carriers can be deep classical. At molecular scale, the modules being coordinated can remain near the quantum-classical boundary. Dimensionality then scales with the coordinated payload, not with carrier frequency itself. The disagreement with the Miller-style picture is therefore not about whether fast bursts matter. It is about *where the dimensionality resides*: in this paper, fast activity can index access or routing, while low-frequency carriers can dominate effective dimensionality by coordinating a larger payload.

The mathematical basis for that claim comes from decoherence geometry. A d-level quantum system occupies a (d^2-1)-dimensional state manifold under the Bures metric [5, 6]. Complete dephasing collapses this manifold onto the $(d-1)$-dimensional classical probability simplex under the Fisher–Rao metric [7, 8]. The collapse is therefore not just loss of "quantumness" in an informal sense; it is a geometric destruction of distinguishable directions in state space.

That perspective is already useful in quantum biology. Photosynthetic complexes are known to operate in regimes where intermediate dephasing can improve transport efficiency [9, 10, 11, 12, 13]. What has been missing is a language for saying how classical or non-classical those biologically relevant operating points actually are. The Bures principal-angle framework supplies that language.

The broader motivation is cognitive. If one assumes that dimensionality is carried mainly by high-frequency activity, then low-frequency waves look low-dimensional by construction. If instead dimensionality is allowed to reside in the coordinated substrate, then the picture changes. A low-frequency wave can be *more* useful precisely because it is more classical: it is a reliable large-scale channel that can entrain many payload modules without introducing much quantum-classical mixing of its own. The core inversion is simple: high frequency may correlate with flexible access, but low frequency may carry more system-level dimensionality because it reaches farther and coordinates broader territories [14].

The paper develops that thesis in four steps. Section 2 states the core geometric facts of decoherence and defines the non-classicality index χ. Section 3 builds a carrier-payload formalism in which local module loads add into a redundancy-adjusted effective payload. Section 4 uses photosynthesis as proof of principle and then prioritizes ion channels as the leading neural payload candidate, with protein microdomains as a secondary extension. Section 5 introduces a minimal phase-window bridge from carrier frequency to coordinated payload size and states the resulting low-frequency/high-dimensional consequence for biological computation. The central outcome is a change of locus: frequency is recast as a coordination parameter, while dimensionality lives in the coordinated payload.

2 Geometry of the Quantum-Classical Boundary

2.1 Dimensional collapse under dephasing

Let $\mathcal{M}_Q$ denote the manifold of $d\times d$ density matrices. For a full-rank system, $\dim \mathcal{M}_Q = d^2-1$: there are $d-1$ independent populations and d^2-d off-diagonal coherence directions. Fix a pointer basis $\{|k\rangle\}_{k=1}^{d}$ and let

$$\Lambda_{\mathrm{dec}}(\rho) = \sum_{k=1}^{d} |k\rangle\langle k|\,\rho\,|k\rangle\langle k| \tag{1}$$

be the complete dephasing channel.

Theorem 2.1 (Dimensional collapse)**.** *Under complete dephasing, the* (d^2-1)*-dimensional quantum manifold collapses onto the* $(d-1)$*-dimensional classical probability simplex:*

$$d^2 - 1 \longrightarrow d - 1. \tag{2}$$

Equivalently, the $d^2 - d$ *off-diagonal coherence directions are annihilated, and the Bures metric on the image of* Λ_{dec} *restricts to the Fisher–Rao metric.*

Proof. The image of Λ_{dec} is the set of diagonal density matrices, which is the classical simplex Δ_{d-1}. Any derivative in an off-diagonal direction vanishes after applying Λ_{dec}, so those directions drop out of the quantum Fisher information matrix. The surviving diagonal block is exactly the classical Fisher information of the populations. Hence only $d-1$ distinguishable directions remain. □

Theorem 2.1 is the geometric core of the paper. It makes decoherence quantitative in the strongest possible sense: decoherence is a rank-reducing projection of state space, not merely a qualitative washing-out of phase information.

2.2 Principal angles and non-classicality

At a diagonal state, diagonal and off-diagonal tangent directions are orthogonal under the Bures metric. Away from the simplex they need not be orthogonal; the failure of orthogonality is exactly what it means for classical and quantum directions to remain mixed.

Definition 2.2 (Non-classicality index)**.** Let $\theta_1, \ldots, \theta_{d-1}$ be the Bures principal angles between the diagonal and off-diagonal tangent subspaces at a full-rank state ρ. Define

$$\chi(\rho) = \frac{1}{d-1} \sum_{a=1}^{d-1} \cos^2 \theta_a. \tag{3}$$

By construction, $\chi(\rho) \in [0, 1]$. Diagonal states have $\theta_a = \pi/2$ for all a, hence $\chi = 0$. Any state with geometric mixing between classical and coherence directions has $\chi > 0$. The index is deliberately modest: it is not an entanglement measure, not a coherence monotone, and not a claim that a system is "quantum" in any maximal sense. It is a scalar summary of how much of the coherence sector remains geometrically accessible.

Remark 2.3. Large dimensionality alone does not imply non-classicality. A very large classical system can still satisfy $\chi = 0$. The claim developed below is therefore extensive rather than ontological: if a system contains many modules with small but nonzero local χ_i, the total payload can scale with system size.

3 Carrier-Payload Architecture

3.1 Local payload load

To move from single systems to biological assemblies, we distinguish the *carrier* from the *payload*. The carrier is the large-scale coordinating process. The payload is the set of local modules whose internal state spaces are being coordinated.

Definition 3.1 (Local non-classical load)**.** For module i with local dimension d_i and non-classicality χ_i, define

$$L_i = \chi_i(d_i^2 - d_i). \tag{4}$$

The term $d_i^2 - d_i$ is the size of the coherence sector that decoherence would destroy under full projection. Multiplying by χ_i gives a geometry-weighted effective load. When $\chi_i = 0$, the module contributes no non-classical payload. When $\chi_i > 0$, some fraction of that coherence sector remains geometrically relevant.

3.2 Redundancy-adjusted effective payload

Biological modules are not independent. If many modules are strongly redundant, naive linear addition overcounts the accessible payload. We therefore carry an explicit redundancy penalty.

Definition 3.2 (Effective payload)**.** Let E be a set of entrained modules and let $\varrho_E \geq 1$ be a redundancy factor. Define

$$\mathcal{L}_{\text{eff}}(E) = \frac{1}{\varrho_E} \sum_{i \in E} L_i \tag{5}$$

and the coordinated dimensionality

$$D_{\text{coord}}(E) = \sum_{i \in E} (d_i - 1) + \mathcal{L}_{\text{eff}}(E). \tag{6}$$

This separates classical capacity from non-classical uplift. The first term counts the surviving simplex dimensions. The second term counts the residual geometrically accessible coherence sector after redundancy compression.

3.3 Extensive payload under nonzero module density

Proposition 3.3 (Extensive non-classical payload)**.** *Let E_N be an entrained set of N modules. Assume:*

(i) *there exist constants $L_\star > 0$ and $p > 0$ such that at least a fraction p of modules in E_N satisfy $L_i \geq L_\star$;*

(ii) *the redundancy factor is uniformly bounded, $\varrho_{E_N} \leq \varrho_\star$.*

Then

$$\mathcal{L}_{\text{eff}}(E_N) \geq \frac{pL_\star}{\varrho_\star} N, \tag{7}$$

and hence the non-classical payload is extensive in N.

Proof. At least pN modules contribute load at least $L_\star$, so $\sum_{i \in E_N} L_i \geq pNL_\star$. Dividing by the bounded redundancy factor gives the stated bound. □

Proposition 3.3 is the paper's main bridge from local molecular geometry to system-scale dimensionality. It does not say that large systems are automatically non-classical. It says that if a nonzero density of modules sits near the boundary, then the total accessible non-classical load grows with system size instead of saturating away.

3.4 Low-frequency advantage as a coordination statement

Corollary 3.4 (Low-frequency carrier advantage)**.** *Let $f_1 < f_2$ be two carrier frequencies with entrained sets $E(f_1)$ and $E(f_2)$. Suppose:*

(i) $|E(f_1)| \geq |E(f_2)|$;

(ii) *the local payload statistics and redundancy bounds remain comparable across the two sets.*

Then

$$D_{\mathrm{coord}}(E(f_1)) \geq D_{\mathrm{coord}}(E(f_2)). \tag{8}$$

Proof. Apply Proposition 3.3 with comparable module statistics and bounded redundancy. The coordinated dimensionality grows with the number of entrained payload modules, not with carrier frequency itself. □

Corollary 3.4 is the formal version of the paper's headline thesis. Low frequency is not valuable because it is intrinsically high-dimensional. It is valuable because it can coordinate a larger ensemble while staying close to a classical carrier limit.

4 Biological Anchors

4.1 Photosynthesis as proof of principle

Photosynthetic exciton transport is the cleanest biological proof of principle because the quantum-to-classical tradeoff is already functionally established [9, 10, 11, 12, 13]. In the reduced rebuilt-repo calculations based on the validated Hamiltonians of Adolphs–Renger and Novoderezhkin [15, 16], the transport optimum occurs in a near-classical regime rather than at maximal coherence. Averaging over the two standard initial-site choices, FMO gives $\gamma_{\mathrm{opt}}/J_{\max} \approx 1.53$ with $\theta_{\min}^{\mathrm{opt}} \approx 87.96°$, while PE545 gives $\gamma_{\mathrm{opt}}/J_{\max} \approx 7.87$ with $\theta_{\min}^{\mathrm{opt}} \approx 89.02°$. At the biological operating point, the rebuilt scans give $\theta_{\min}^{\mathrm{bio}} \approx 87.67°$ (FMO) and $89.02°$ (PE545). In local-load units $L = \chi(d^2 - d)$, those same biological points correspond to $L_{\mathrm{FMO}}^{\mathrm{bio}} \approx 2.28 \times 10^{-2}$ and $L_{\mathrm{PE545}}^{\mathrm{bio}} \approx 4.93 \times 10^{-3}$. This is exactly the regime the present framework is built to describe: biologically relevant function with $\chi > 0$ but principal angles already close to $\pi/2$.

The role of photosynthesis in this paper is strategic. It shows that the near-boundary regime is not empty rhetoric. Biology can already be shown to function there. That makes it plausible that other molecular biological systems may also carry small but nonzero local loads. Table 1 and Figure 1 summarize the current anchor set and, just as importantly, show the split the paper needs: rebuilt molecular payload anchors cluster in the same near-boundary band as the rebuilt photosynthetic proof-of-principle calculations, while the rebuilt neural carrier proxy already sits farther right and closer to the classical limit. The comparison is not only angular: the ion-channel local load $L_{\mathrm{ion}} \approx 3.31 \times 10^{-3}$ is of the same order as PE545 at the biological point, so the neural payload anchor is not being placed in a wholly alien regime.

The table is doing more work than a simple operating-point list. Its χ column makes the carrier-payload separation explicit: payload anchors cluster in a small-but-nonzero non-classical band around 10^{-4}, while the rebuilt gamma-carrier proxy is already down at 3.72×10^{-6} by $\gamma/J_{\max} = 50$. The paper's central split is therefore not rhetorical; it is numerically visible in the anchor comparison itself.

4.2 Ion channels as the primary payload example

If the paper is meant to say something concrete about cognition, ion channels come first. They sit at the molecular interface between microscopic state transitions and population-level excitability; they exist in enormous copy number; and they offer a direct route from local non-classicality to large entrained site counts. Structurally, selectivity filters and gating assemblies

System	Role	Src.	$\gamma_{\text{bio}}/J_{\max}$	$\theta_{\min}^{\text{bio}}$	χ^{bio}	Use in draft
FMO	Proof point	Fresh	1.14	87.67°	5.42×10^{-4}	Functional near-boundary photosynthesis anchor.
PE545	Proof point	Fresh	7.88	89.02°	8.80×10^{-5}	Independent photosynthetic confirmation of the same band.
Ion channel	Neural payload	Fresh	6.67	88.55°	2.76×10^{-4}	Fresh molecular anchor for the primary neural payload route.
Prot. microd.	Payload ext.	Fresh	7.69	88.21°	3.66×10^{-4}	Fresh secondary payload anchor outside ion conduction.
Neural oscillation	Carrier	Proxy	50	89.78°	3.72×10^{-6}	Rebuilt gamma-band carrier proxy, already near classical.

Table 1: Anchor systems used in the present draft. All rows are computed in the rebuilt repo. The photosynthetic models are reduced to the quantities this draft actually needs: biological-point and transport-optimum geometry from validated Hamiltonians. The neural row reports the rebuilt carrier proxy at $\gamma/J_{\max} = 50$ rather than the true biological limit $\gamma_{\text{bio}}/J_{\max} \gg 1$; the latter only pushes the carrier further toward the classical limit.

are small-dimensional molecular machines with repeated motifs, constrained geometries, and biologically relevant couplings [17, 18]. In the present framework that makes them the cleanest neural payload candidate: small local d, plausible boundary-near energy scales, and a direct path to large N.

To keep the rebuilt repo honest, we computed a minimal 4-site KcsA-like selectivity-filter anchor directly in the fresh codebase. Using site energies $(30, 45, 45, 30)\,\text{cm}^{-1}$, nearest-neighbor coupling $J_{\max} = 30\,\text{cm}^{-1}$, biological dephasing $\gamma_{\text{bio}} = 200\,\text{cm}^{-1}$, and trapping from S_1 to S_4, the biological operating point sits at $\gamma_{\text{bio}}/J_{\max} = 6.67$ with

$$\theta_{\min} \approx 88.55^\circ, \qquad \chi \approx 2.76 \times 10^{-4}. \tag{9}$$

The full principal-angle spectrum at the biological point is (88.55°, 89.22°, 89.91°). This is exactly the regime the paper needs: not strongly quantum, not fully classical, but slightly non-classical in a geometrically explicit sense.

The corresponding bounded transport scan in this simple trap model is edge-peaked rather than ENAQT-like, so transport optimization is *not* the claim being made for ion channels here. The role of the model is narrower: it establishes a concrete boundary-near neural payload anchor in the rebuilt repo. That is exactly the regime needed by Proposition 3.3. If each channel carries only a very small local load, the aggregate payload can still be large whenever many channels are entrained and redundancy is not overwhelming. For the neural application, this is the most direct route from local geometry to system-level dimensionality.

A compact robustness sweep makes the same point without overselling the model. Varying the nearest-neighbor coupling over $J_{\max} \in \{15, 20, 30, 40\}\,\text{cm}^{-1}$ and the dephasing over $\gamma \in \{100, 150, 200, 250, 300\}\,\text{cm}^{-1}$ keeps the biological-readout geometry in the same band: across the entire 20-point grid, $\theta_{\min}$ stays between 86.43° and 88.93°, while χ stays between 1.53×10^{-4} and 1.55×10^{-3}. Weaker coupling pushes the payload somewhat farther from the classical simplex; stronger coupling pulls it closer. But the sweep does not produce a collapse into either a strongly quantum or exactly classical regime. For the present thesis, that is the right robustness statement: the ion-channel payload anchor is parameter-sensitive, but not a one-point artifact.

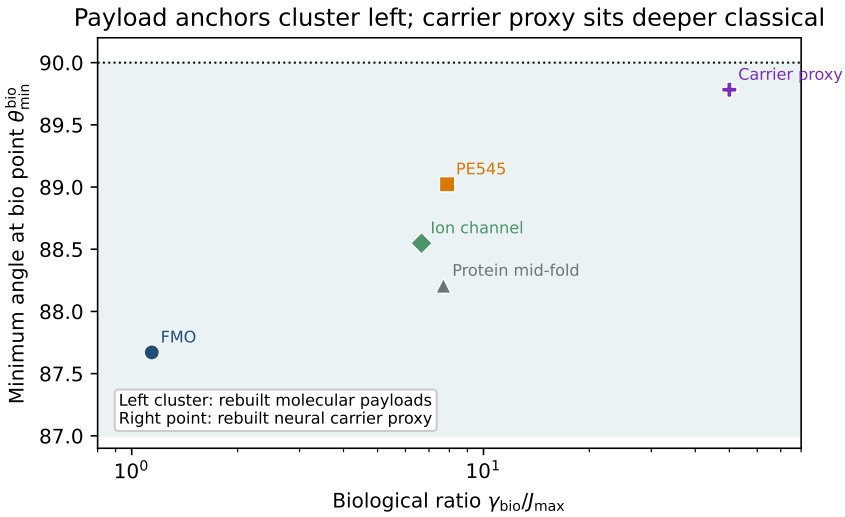

Figure 1: Biological operating points for the rebuilt anchor set. The molecular payload anchors — FMO, PE545, the ion-channel point, and the protein mid-fold point — cluster in the same near-boundary band, with $\theta_{\min}^{\text{bio}}$ between roughly 87.7° and 89.0°. The rebuilt neural carrier proxy sits farther right at $\gamma/J_{\max} = 50$ and closer to the classical limit. The visual split is therefore the paper's central one: functional or candidate payload systems remain small-but-nonzero in χ, while the carrier is already substantially more classical on the same geometry map.

4.3 Protein microdomains as secondary payloads

Protein-folding microdomains, ligand-binding pockets, and other conformational substructures remain secondary in this paper, but they are now secondary with an actual rebuilt anchor rather than only prose. Using the illustrative $d = 6$ mid-fold network recovered from the earlier checkpoint and recomputed in the fresh pipeline,

$$H_{\text{fold}} = \begin{pmatrix} 20 & 25 & 12 & 0 & 0 & 0 \\ 25 & 35 & 18 & 22 & 0 & 0 \\ 12 & 18 & 42 & 20 & 10 & 0 \\ 0 & 22 & 20 & 50 & 24 & 12 \\ 0 & 0 & 10 & 24 & 38 & 26 \\ 0 & 0 & 0 & 12 & 26 & 28 \end{pmatrix}, \tag{10}$$

with initial site 0, target site 5, $\kappa = 0.5\,\text{ps}^{-1}$, and biological dephasing $\gamma_{\text{bio}} = 200\,\text{cm}^{-1}$, we obtain $J_{\max} = 26\,\text{cm}^{-1}$ and therefore $\gamma_{\text{bio}}/J_{\max} \approx 7.69$. At the biological point the same Lindblad-plus-geometry pipeline gives $\theta_{\min} \approx 88.21°$ and $\chi \approx 3.66 \times 10^{-4}$, with a local load $L_{\text{fold}} \approx 1.10 \times 10^{-2}$. So the folding microdomain lands in the same near-classical but non-classical region as the ion-channel payload. That broadens the payload story beyond excitability while preserving the present paper's hierarchy: ion channels remain the primary neural route, protein microdomains are the secondary generalization.

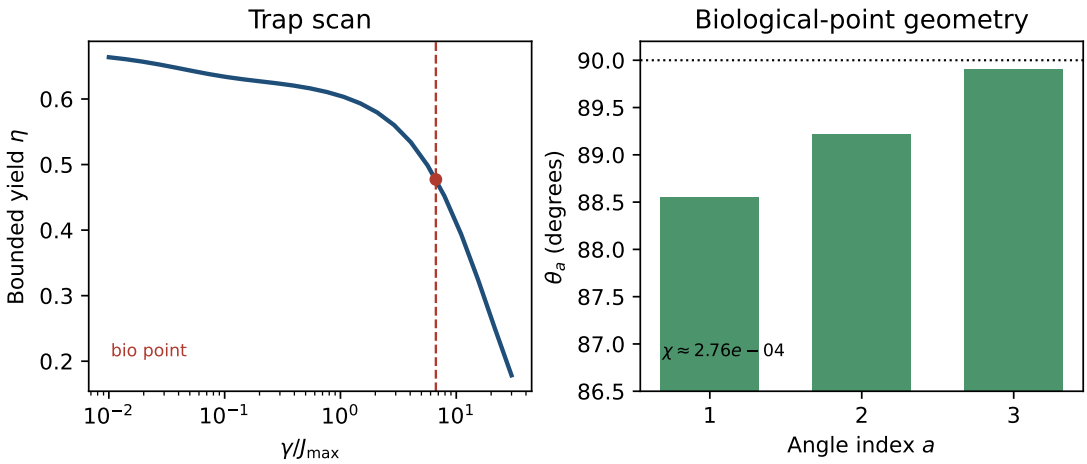

Figure 2: Minimal ion-channel payload anchor from the rebuilt repo. Left: bounded trapping yield for the KcsA-like 4-site model as a function of $\gamma/J_{\max}$, with the biological point marked. Right: the three Bures principal angles at the biological point. The left panel is shown only to locate the modeled operating range; because the bounded yield is edge-peaked in this simple trap model, it is not used as an ENAQT claim. The geometric result is the point of the figure: at $\gamma_{\text{bio}}/J_{\max} = 6.67$, the ion-channel payload remains boundary-near with $\theta_{\min} \approx 88.55°$ and $\chi \approx 2.76 \times 10^{-4}$.

5 Low-Frequency Coordination and Intelligence

5.1 Classical carriers, non-classical payloads

Neural oscillations at behavioral timescales are expected to be deep classical carriers. At those scales, dephasing dominates so strongly that the carrier itself should have $\chi_{\text{carrier}} \approx 0$. A reduced rebuilt-repo gamma-band proxy confirms the direction of that limit: for a synthetic $d = 10$ microcircuit, $\theta_{\min} \approx 89.43°$ with $\chi \approx 3.29 \times 10^{-5}$ already at $\gamma/J_{\max} = 1$, and $\theta_{\min} \approx 89.78°$ with $\chi \approx 3.72 \times 10^{-6}$ by the proxy point $\gamma/J_{\max} = 50$. Even this conservative proxy puts the carrier roughly 74× lower in χ than the ion-channel payload anchor at its biological point. This is not a defect in the argument; it is the feature that makes the carrier useful. A deep-classical carrier can propagate, synchronize, and entrain without adding significant geometric mixing of its own. That is why low-frequency waves are candidates for large-scale control rather than obstacles to complexity.

The disagreement with the Miller-style high-frequency/high-dimensional intuition can now be stated cleanly. High-frequency bursts may indeed track local access or flexible routing [1, 2]. This paper does not deny that. It denies the stronger inference that frequency itself is where dimensionality lives. Dimensionality resides in the coordinated substrate. If lower-frequency waves entrain larger cortical territories and therefore larger ion-channel payload sets [14], then Corollary 3.4 implies a higher coordinated dimensionality even when the carrier is more classical.

5.2 What "low frequency = high dimensional" means

The slogan needs to be read precisely. It does *not* mean that a low-frequency oscillation is internally more complicated than a high-frequency oscillation. It means:

$$f \downarrow \implies |E(f)| \uparrow \implies D_{\text{coord}}(E(f)) \uparrow, \tag{11}$$

under the stated payload and redundancy assumptions.

This reframes the intelligence question. If cognition depends on the coordinated manipulation of many mixed-selectivity degrees of freedom [3, 4], then one should look for the geometry of the coordinated substrate, not just the spectral properties of the carrier. Low-frequency activity can be the right place to look for intelligence not because it is itself non-classical, but because it may be the most effective large-scale organizer of an ion-channel-rich payload that is not fully classical. Protein microdomains extend the same logic beyond excitability, but they are secondary to the ion-channel route for the present argument.

5.3 A minimal phase-window carrier model

The remaining question is how frequency enters the picture dynamically rather than rhetorically. A more physical bridge than the pure threshold model is a phase-window carrier model. Let a carrier of frequency f propagate with speed v and amplitude envelope

$$A(r) = A_0 \exp\left(-\frac{r}{\lambda}\right), \tag{12}$$

where r is distance from the source and λ is an amplitude attenuation length. A payload module at distance r is coordinatable only if two conditions both hold:

(i) the carrier amplitude exceeds threshold, $A(r) \geq A_{\text{thr}}$;

(ii) the propagation-induced phase lag remains inside a tolerable phase window, $2\pi f r/v \leq \Delta\phi_{\max}$.

These two constraints define an amplitude-limited radius

$$R_{\text{amp}} = \lambda \log\left(\frac{A_0}{A_{\text{thr}}}\right), \tag{13}$$

and a phase-limited radius

$$R_\phi(f) = \frac{v\,\Delta\phi_{\max}}{2\pi f}. \tag{14}$$

The effective entrained radius is then

$$R_{\text{eff}}(f) = \min\big(R_{\text{amp}}, R_\phi(f)\big). \tag{15}$$

If payload modules populate an effective spatial support of dimension s with density ρ_m, then

$$N(f) = |E(f)| \approx \rho_m C_s R_{\text{eff}}(f)^s = \rho_m C_s \min\left(R_{\text{amp}}, \frac{v\,\Delta\phi_{\max}}{2\pi f}\right)^s, \tag{16}$$

where C_s is the unit-ball volume factor in dimension s. In the phase-limited regime $R_\phi(f) \leq R_{\text{amp}}$, this gives $N(f) \propto f^{-s}$; at sufficiently low frequency the growth saturates at the amplitude ceiling R_{amp}.

Proposition 5.1 (Phase-window scaling with saturation)**.** *Assume the phase-window model above, a common payload dimension d_m, and the nonzero-density/bounded-redundancy conditions of Proposition 3.3. Then*

$$D_{\text{coord}}(E(f)) \geq \rho_m C_s \min\left(R_{\text{amp}}, \frac{v\,\Delta\phi_{\max}}{2\pi f}\right)^s \left[(d_m - 1) + \frac{pL_\star}{\varrho_\star}\right] \tag{17}$$

In the phase-limited regime, coordinated dimensionality grows as f^{-s} as frequency falls; in the amplitude-limited regime, the growth saturates. In both cases, the low-frequency advantage comes from coordinating a larger payload set, not from the carrier itself becoming less classical.

Proof. Equation (16) gives the size of the entrained set under the phase-window model. For modules of common dimension d_m, Definition 3.2 gives

$$D_{\text{coord}}(E(f)) = N(f)(d_m - 1) + \mathcal{L}_{\text{eff}}(E(f)).$$

Applying Proposition 3.3 to the entrained set yields

$$\mathcal{L}_{\text{eff}}(E(f)) \geq \frac{pL_\star}{\varrho_\star} N(f),$$

which gives the stated bound after substitution. □

Proposition 5.1 is still only a toy bridge. It does not fit real cortical attenuation, conduction anisotropy, or laminar structure. Its role is narrower and sufficient: it makes the paper's inversion explicit in one model that is at least tied to propagation delay and phase tolerance. Frequency enters through the admissible phase-lag radius. Dimensionality enters through the payload geometry being coordinated.

A reasonable objection is that the hard boundary in (16) could be generating the low-frequency advantage by construction. To check that, Figure 3 also overlays a heterogeneous payload-field simulation. We distribute 12,000 payload modules across a 2D disc of radius $6R_0$, assign lognormally distributed amplitude ceilings $R_{\text{amp},i}$ around $4R_0$ and baseline phase radii $R_{\phi,i}(f_0)$ around R_0 (log-width 0.35), and count module i as entrained when both constraints are satisfied at frequency f. The simulation is normalized so that $N(f_0) = N_0$. The resulting payload load tracks the analytic bridge near $f \sim f_0$, remains monotone as frequency falls, and still gives an approximately 2.8×10^2 gain in coordinated payload between $f/f_0 = 4$ and 0.25 at $\varrho = 1$ over the plotted range. So the inversion is not an artifact of a single sharp threshold: it survives distributed tolerances and finite support.

Using the rebuilt-repo ion-channel anchor makes the scale of the effect explicit. For the 4-site selectivity-filter model,

$$L_{\text{ion}} = \chi(d^2 - d) \approx (2.76 \times 10^{-4}) \times 12 \approx 3.31 \times 10^{-3} \tag{18}$$

per channel at the biological point. If one takes an illustrative baseline of $N_0 = 10^4$ entrained channels at $f = f_0$ in the phase-limited regime, then the phase-window bridge gives

$$\mathcal{L}_{\text{eff}}(f) \approx \frac{N_0 L_{\text{ion}}}{\varrho} \min\left(\frac{R_{\text{amp}}}{R_0}, \frac{f_0}{f}\right)^s, \qquad R_0 := R_\phi(f_0). \tag{19}$$

Figure 3 plots this for $s = 2$, $R_{\text{amp}}/R_0 = 4$, and redundancy factors $\varrho \in \{1, 10, 100\}$. The point is not the exact numerical choice of N_0 or R_{amp}/R_0; it is that a very small local non-classical load can become system-relevant once multiplied by large entrained site counts, while the low-frequency gain saturates rather than diverging forever.

There is also a clean way to state the fractional size of the effect. For homogeneous payload modules,

$$\Phi := \frac{\mathcal{L}_{\text{eff}}}{N(d_m - 1)} = \frac{\chi(d_m^2 - d_m)}{\varrho(d_m - 1)} = \frac{\chi d_m}{\varrho}. \tag{20}$$

N	Classical $3N$	$\mathcal{L}_{\text{eff}}$ $(\varrho = 1)$	$\mathcal{L}_{\text{eff}}$ $(\varrho = 10)$	$\mathcal{L}_{\text{eff}}$ $(\varrho = 100)$
10^3	3.0×10^3	3.31	3.31×10^{-1}	3.31×10^{-2}
10^4	3.0×10^4	3.31×10^1	3.31	3.31×10^{-1}
10^5	3.0×10^5	3.31×10^2	3.31×10^1	3.31

Table 2: Ion-channel payload benchmarks at the biological point, using the rebuilt-repo anchor $L_{\text{ion}} \approx 3.31 \times 10^{-3}$ per channel. The corresponding fractional uplifts above the classical baseline are $\Phi_{\text{ion}} \approx 1.10 \times 10^{-3}$ for $\varrho = 1$, 1.10×10^{-4} for $\varrho = 10$, and 1.10×10^{-5} for $\varrho = 100$. The message is not that the fraction is large; it is that the absolute non-classical payload remains extensive in N.

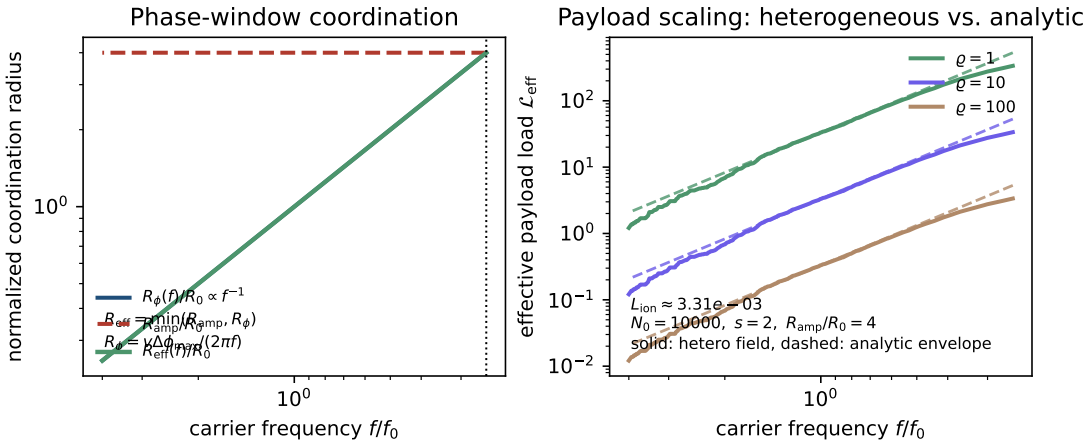

Figure 3: Phase-window bridge used in Section 5. Left: the coordination radius is limited by the smaller of two scales, the amplitude ceiling R_{amp} and the phase-lag radius $R_\phi(f) = v\Delta\phi_{\max}/(2\pi f)$. Lower-frequency carriers enlarge the phase-limited radius until the amplitude ceiling is reached. Right: using the actual ion-channel anchor from Figure 2, dashed curves show the analytic phase-window payload load $\mathcal{L}_{\text{eff}}(f) \approx N_0 L_{\text{ion}} \min(R_{\text{amp}}/R_0, f_0/f)^s/\varrho$ for illustrative $N_0 = 10^4$, $s = 2$, $R_{\text{amp}}/R_0 = 4$, and redundancy factors $\varrho \in \{1, 10, 100\}$. Solid curves show a heterogeneous payload-field simulation with distributed amplitude ceilings and phase tolerances, normalized to match $N(f_0) = N_0$. The heterogeneous bridge preserves the same low-frequency advantage and saturation structure, so the effect is not created by a single hard threshold.

Thus the aggregate non-classical payload is extensive in N, but the fractional uplift above the classical baseline is independent of N for fixed (d_m, χ, ϱ). For the ion-channel anchor ($d_m = 4$, $\chi \approx 2.76 \times 10^{-4}$),

$$\Phi_{\text{ion}} \approx \frac{1.10 \times 10^{-3}}{\varrho}, \tag{21}$$

so the toy bridge says two things simultaneously: the per-channel effect is small, and the absolute payload load still scales linearly with the number of entrained channels.

5.4 Predictions

The carrier-payload view yields direct predictions:

1. Measures of cognitive dimensionality should track entrained ion-channel payload diversity better than carrier frequency alone.

2. Manipulations that preserve large-scale oscillatory coherence but reduce the diversity or number of available molecular payload sites should reduce effective dimensionality.

Observable or manipulation	Frequency-as-dimensionality expectation	Carrier-payload expectation
Cross-condition variation in measured dimensionality	Dimensionality should increase primarily with carrier frequency or burstiness.	Dimensionality should increase primarily with entrained payload size and payload diversity, even when the carrier is slower.
Hold large-scale carrier coherence fixed while reducing available payload sites or payload diversity	Only modest changes in dimensionality are expected if the carrier spectrum is preserved.	Effective dimensionality should fall because the payload, not the carrier frequency, is where state-space richness resides.
Compare low-frequency broad-coordination states with high-frequency local-access states	High-frequency states should appear more dimensional because they are spectrally faster.	Low-frequency states can appear more dimensional when they coordinate a larger payload ensemble, even if high-frequency states remain better markers of local access or routing.

Table 3: Rival empirical expectations. The paper does not deny that high-frequency bursts can correlate with flexible access or gating. It denies the stronger inference that carrier frequency itself is the primary locus of dimensionality.

3. If low-frequency rhythms have larger phase-tolerant coordination radii, then the coordinated dimensionality should scale with entrained territory size as in Proposition 5.1, not with frequency alone.

4. The right computational bottleneck is not whether classical surrogates exist in principle, but how expensive it is to preserve the effective payload $\mathcal{L}_{\text{eff}}$ rather than projecting it away, even when the fractional uplift Φ is modest.

These predictions are useful only if they are contrasted with the position the paper is trying to replace. Table 3 states that contrast in the sharpest operational form available in the present draft.

6 Discussion

The main gain of the present reformulation is conceptual clarity. The old binary question—is biology quantum or classical?—is the wrong one. The geometric question is better: how much of the coherence sector remains accessible, where is it located, and how is it coordinated across scales?

Three claims survive scrutiny in the current draft. First, decoherence is a dimensional projection of state space. Second, near-boundary biological dynamics are real, with photosynthesis as the proof of principle. Third, low-frequency carrier advantage is mathematically coherent once dimensionality is relocated to the coordinated payload and coupled to frequency through a transparent phase-window toy model.

Several things remain open. The rebuilt repo now has two neural-relevant payload anchors, but the molecular payload classes beyond photosynthesis, ion channels, and protein microdomains still need fresh computations. The redundancy factor ϱ_E needs empirical grounding rather than remaining a formal placeholder. And the neuroscience side still needs a biophysically serious carrier-to-payload coupling model beyond the reduced phase-window bridge introduced

in Section 5. Those are real gaps. They are also the right gaps: they are downstream of a coherent theory rather than upstream of one.

It is also worth stating plainly what the rebuilt repo does and does not yet contain. It now computes the reduced photosynthetic proof-of-principle anchors, two concrete neural-side payload anchors directly, and a reduced neural carrier proxy. What it does *not* yet contain is the full old figure stack or the broader family of molecular payload models beyond ion channels and protein microdomains. That is the right asymmetry for the present thesis-first draft: only the computations that sharpen the carrier-payload argument have been brought forward.

The paper therefore makes a narrower but more defensible claim than strong quantum-cognition proposals. It does not say that neural oscillations are quantum. It does not say that high dimensionality forces non-classicality. It says that biological systems built from many structured molecular machines can carry an extensive non-classical payload if a nonzero density of those machines sits near the quantum-classical boundary. Large-scale low-frequency waves may then be the right place to look for intelligence because they are the best classical carriers of that payload. In neural tissue, ion channels are the first place to test that claim. The numerical lesson of the rebuilt draft is correspondingly sober: the local fractional uplift is small [Eq. (21)], but the absolute payload remains extensive in entrained site count.

That is the full intended inversion of the usual intuition. Frequency is not dimensionality. Frequency is coordination. Dimensionality lives in the substrate being coordinated.

References

[1] Mikael Lundqvist, Josh Rose, Pawel Herman, Scott L. Brincat, Timothy J. Buschman, and Earl K. Miller. Gamma and beta bursts underlie working memory. *Neuron*, 90(1):152–164, 2016.

[2] Earl K. Miller, Mikael Lundqvist, and André M. Bastos. Working memory 2.0. *Neuron*, 100(2):463–475, 2018.

[3] Mattia Rigotti, Omri Barak, Melissa R. Warden, Xiao-Jing Wang, Nathaniel D. Daw, Earl K. Miller, and Stefano Fusi. The importance of mixed selectivity in complex cognitive tasks. *Nature*, 497(7451):585–590, 2013.

[4] Stefano Fusi, Earl K. Miller, and Mattia Rigotti. Why neurons mix: High dimensionality for higher cognition. *Current Opinion in Neurobiology*, 37:66–74, 2016.

[5] Samuel L. Braunstein and Carlton M. Caves. Statistical distance and the geometry of quantum states. *Physical Review Letters*, 72(22):3439–3443, 1994.

[6] Dénes Petz and Csaba Sudár. Geometry of quantum states. *Journal of Mathematical Physics*, 37(6):2662–2673, 1996.

[7] C. R. Rao. Information and the accuracy attainable in the estimation of statistical parameters. *Bulletin of the Calcutta Mathematical Society*, 37:81–91, 1945.

[8] Shun-ichi Amari. *Differential-Geometrical Methods in Statistics*. Springer, 1985.

[9] Gregory S. Engel, Tessa R. Calhoun, Elizabeth L. Read, Tae-Kyu Ahn, Tomas Mancal, Yuan-Chung Cheng, Robert E. Blankenship, and Graham R. Fleming. Evidence for wavelike energy transfer through quantum coherence in photosynthetic systems. *Nature*, 446(7137):782–786, 2007.

[10] Masoud Mohseni, Patrick Rebentrost, Seth Lloyd, and Alán Aspuru-Guzik. Environment-assisted quantum walks in photosynthetic energy transfer. *The Journal of Chemical Physics*, 129(17):174106, 2008.

[11] Martin B. Plenio and Susana F. Huelga. Dephasing-assisted transport: Quantum networks and biomolecules. *New Journal of Physics*, 10(11):113019, 2008.

[12] Patrick Rebentrost, Masoud Mohseni, Ivan Kassal, Seth Lloyd, and Alán Aspuru-Guzik. Environment-assisted quantum transport. *New Journal of Physics*, 11(3):033003, 2009.

[13] Jianshu Cao, Richard J. Cogdell, David F. Coker, Hong-Guang Duan, Jan Hauer, Ulrich Kleinekathöfer, Thomas L. C. Jansen, Tomáš Mančal, R. J. Dwayne Miller, Jennifer P. Ogilvie, Valeria I. Prokhorenko, Thomas Renger, Huan-Qiang Tan, Roel Tempelaar, Michael Thorwart, Erling Thyrhaug, Sebastian Westenhoff, and Donatas Zigmantas. Quantum biology revisited. *Science Advances*, 6(14):eaaz4888, 2020.

[14] György Buzsáki and Xiao-Jing Wang. Mechanisms of gamma oscillations. *Annual Review of Neuroscience*, 35:203–225, 2012.

[15] Jörg Adolphs and Thomas Renger. How proteins trigger excitation energy transfer in the FMO complex of green sulfur bacteria. *Biophysical Journal*, 91(8):2778–2797, 2006.

[16] Vladimir I. Novoderezhkin, Alexander B. Doust, Carles Curutchet, Gregory D. Scholes, and Rienk van Grondelle. Excitation dynamics in phycoerythrin 545: Modeling of steady-state spectra and transient absorption with modified Redfield theory. *Biophysical Journal*, 99(2):344–352, 2010.

[17] Declan A. Doyle, João M. Cabral, Rolf A. Pfuetzner, Anling Kuo, Juris M. Gulbis, Sandra L. Cohen, Brian T. Chait, and Roderick MacKinnon. The structure of the potassium channel: Molecular basis of K^+ conduction and selectivity. *Science*, 280(5360):69–77, 1998.

[18] Bertil Hille. *Ion Channels of Excitable Membranes*. Sinauer Associates, 3 edition, 2001.